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(1) Research Program/Project/Study Title: <i>Post-harvest Behaviour of Organically Grown Kangkong Applied with Different Soil Ameliorants</i>	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Vegetable	(8) Discipline/Field of Specialization Agriculture

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The experiment was conducted at the Crop and Soil Science Laboratory of Capiz State University - Burias Campus, Burias, Mambusao, Capiz. It was designed as a single-factor experiment laid out in a Completely Randomized Design (CRD) with three replications. The treatments consisted of different soil ameliorants, namely: T1 - Vermicast, T2 - Vermitea, and T3 - Ordinary Compost. The study recorded the visual quality rating (VQR), shriveling rate, and chemical analysis. The data collected were subjected to the Analysis of Variance (ANOVA) and F-test for Factorial CRD using STAR software. Mean differences among treatments were determined using the Least Significant Difference (LSD).

The result of the study showed that all the parameters of organically grown kangkong was not affected by the different soil ameliorants applied. The visual quality rating (VQR) of kangkong remained consistently high but declined over time. There is a gradual increase in shriveling of kangkong from the fourth day to the sixth day of storage for all treatments. The pH levels of kangkong decreased over time regardless of the soil ameliorants applied. The initial and final total soluble solids (TSS) of kangkong were not affected by the application of the different soil ameliorants.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)
organic farming, soil ameliorants, vermicompost, compost, kangkong

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Kangkong (*Ipomoea aquatica*), is a popular leafy vegetable cultivated for its tender shoots and leaves, which are widely consumed in various culinary dishes. The increasing demand for organic produce and sustainable farming practices has prompted the cultivation of organically grown kangkong, driven by the desire to reduce synthetic chemical inputs, promote environmental sustainability, and ensure safe and nutritious food for consumers (Ecoport, 2016). Organic kangkong production aims to enhance soil health, minimize environmental impact, and provide a healthier and more environmentally responsible food choice.

The post-harvest phase is a pivotal component of kangkong production, significantly affecting the overall quality, shelf life, and market value of the crop. Post-harvest losses can occur due to various factors, including mechanical damage, pathogen contamination, and unfavorable storage conditions (Bhargava et al., 2015). To mitigate these challenges and improve the post-harvest behavior of organically grown kangkong, it is imperative to investigate the potential benefits of different soil ameliorants applied during the cultivation process.

Soil ameliorants, including organic matter and mineral amendments, have been widely recognized for their ability to enhance soil fertility, structure, and overall health (Zotarelli et al., 2018). The application of soil ameliorants can influence root development, nutrient uptake, and overall crop performance. Previous research has

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demonstrated the positive effects of soil ameliorants on the yield and quality of various crops (Manojlovic et al., 2018). However, there is a need to investigate how different soil ameliorants can impact the post-harvest behavior of organically grown kangkong.

The study aimed to determine the post-harvest behavior of organically grown kangkong applied with different soil ameliorants, in terms of visual quality rating, shriveling rate, pH levels, and total soluble solids. By exploring the intricate relationships between soil ameliorants and the post-harvest behavior of kangkong, this research contributes to the development of sustainable and environmentally friendly practices for kangkong production, benefiting both growers and consumers.

Objectives of the Study

In general, the purpose of this study was to determine the post-harvest behaviour of organically grown kangkong applied with different soil ameliorants. Specifically, the study aimed to;

1. evaluate the visual quality rating of organically grown kangkong after post-harvest storage;
2. determine the shriveling rate of kangkong when subjected to different soil ameliorants during post-harvest storage;
3. measure the changes in pH levels of kangkong after post-harvest storage when treated with different soil ameliorants;
4. quantify the total soluble solids (TSS) content in organically grown kangkong following post-harvest storage with varying soil ameliorants.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The experiment was conducted at the Crop and Soil Science Laboratory of Capiz State University - Burias Campus, Burias, Mambusao, Capiz. It was designed as a single-factor experiment laid out in a Completely Randomized Design (CRD) with three replications. The treatments consisted of different soil ameliorants, namely: T1 - Vermicast, T2 - Vermitea, and T3 - Ordinary Compost. The study recorded the visual quality rating (VQR), shriveling rate, and chemical analysis. The data collected were subjected to the Analysis of Variance (ANOVA) and F-test for Factorial CRD using STAR software. Mean differences among treatments were determined using the Least Significant Difference (LSD).

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Visual Quality Rating

The results showed that kangkong, regardless of the soil ameliorants applied, received a score of 9 during the second day of storage, indicating excellent, field-fresh

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produce with no defects. However, a decline in VQR was observed from the fourth day of storage until the sixth day. Despite this decline, there was no significant difference observed among treatments. Nevertheless, it was noted that the application of Vermitea on the kangkong had a similar effect as Vermicast and ordinary compost.

The maintenance of consistently high VQR is pivotal for preserving the economic value and consumer satisfaction associated with kangkong consumption. However, it is widely recognized that the VQR of kangkong may experience a decline over time, raising concerns about post-harvest handling, storage, and quality preservation (Rajasekaran et al., 2019). The VQR of kangkong tends to gradually decline as it undergoes storage conditions, reflecting the natural aging and degradation processes that occur in harvested vegetables (Kheng et al., 2016). These changes can encompass alterations in color, wilting, and physical damage, all of which can negatively affect the visual appeal of kangkong and, consequently, its marketability.

Shriveling Rate

During the storage of kangkong, the shriveling rate gradually increased from the fourth to the sixth day, regardless of the soil ameliorants applied. It was observed that there was no difference in the shriveling rate of kangkong despite applying different soil ameliorants. On the fourth day of storage, the shriveling rate reached level 3, indicating that 26-50% of the crop was moderately affected by shriveling. On the sixth day, the shriveling rate reached level 4, indicating that 50% of the kangkong area was severely affected by shriveling.

However, the ANOVA revealed that the different soil ameliorants did not have a significant effect on the shriveling rate of kangkong. This suggests that the application of different soil ameliorants did not cause any change in the shriveling rate of kangkong.

During storage, kangkong continues to respire and lose water, resulting in the reduction of turgor pressure and the development of shriveled, wilted leaves. The rate of moisture loss can be influenced by storage conditions, temperature, humidity, and the application of different treatments or soil ameliorants during cultivation (Sarker et al., 2019). In many cases, kangkong may experience a gradual increase in shriveling during the post-harvest period (Kader, 2002). These changes can have implications for the overall appearance and marketability of kangkong.

Initial pH and Final pH

The results of the study revealed that the initial pH level of kangkong applied with Vermicast was the highest at 6.34, followed by radish applied with ordinary compost at 6.30. The lowest initial pH level was found in radish treated with Vermitea at 6.25. Moreover, the final pH level of kangkong applied with Vermitea was the highest at 5.99, followed by kangkong applied with Vermicast at 5.90. The lowest was observed for kangkong applied with ordinary compost, with an average of 5.88. The pH level of kangkong decreased over time from the initial measurement to the final

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day of storage. However, the Analysis of Variance indicated that the application of soil ameliorants did not have a significant impact on the initial and final pH levels of kangkong.

The pH of kangkong is influenced by various factors, including the natural acidity or alkalinity of the vegetable and changes in organic acids during respiration and other metabolic processes. During the post-harvest period, the pH levels of kangkong may shift due to ongoing biochemical reactions. These changes can affect the overall quality, taste, and texture of the vegetable (Hussain et al., 2017).

Initial TSS and Final TSS

The results of the ANOVA suggest that the use of different soil ameliorants did not have a significant impact on the Total Soluble Solids (TSS) of kangkong. This indicates that the parameter was not influenced by the various treatments administered. The mean initial TSS values ranged from 1.27 to 1.90, while the mean final TSS values ranged from 2.73 to 2.90.

TSS levels can naturally vary during the post-harvest phase due to respiration, enzymatic activity, and other metabolic processes that can impact the overall quality of kangkong (Yang et al., 2018). In certain cases, the application of different soil ameliorants may not significantly impact the initial and final TSS values of the harvested kangkong (Mikkelsen et al., 2016).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

Based on the results of the study, the following conclusions were drawn;

1. The visual quality rating (VQR) of kangkong remained consistently high but declined over time.
2. There is a gradual increase in shriveling of kangkong from the fourth day to the sixth day of storage for all treatments.
3. The pH levels of kangkong decreased over time regardless of the soil ameliorants applied.
4. The initial and final total soluble solids (TSS) of kangkong were not affected by the application of the different soil ameliorants.

Recommendation

Based on the results of the study, the following conclusions were drawn;

1. Use any of the soil ameliorants in growing kangkong organically.

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2. Conduct studies considering the following:

- storage practices to maintain visual quality over time, such as improved packaging and temperature control.
- optimizing storage conditions and handling practices to prevent shriveling, particularly from the fourth to the sixth day of storage.
- methods to stabilize pH during storage of different crops.
- develop strategies to maintain the shriveling rate in storage of different crops.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.

Bhargava, A., Srivastava, S., Singh, R., & Singh, R. B. (2015). Sustainable management of post-harvest diseases of fruits and vegetables: A review. *International Journal of Food, Agriculture & Environment*, 13(2), 295-301.

Ecoport. (2016). *Ipomoea aquatica*. Retrieved from http://ecoport.org/ep?Plant=37139&entityType=PL****&entityDisplayCategory=full

Kader, A. A. (2002). *Postharvest technology of horticultural crops*. University of California, Agriculture and Natural Resources.

Kheng, L. W., Hasan, R., Subramaniam, K., & Chen, C. K. (2016). Sensory attributes, biochemical properties, and consumer acceptance of organically grown tomato (*Solanum lycopersicum* L.). *Journal of Food Quality*, 39(5), 534-546.

Manojlovic, M., Lignier, V., Le Bon, C., Savary-Auzeloux, I., & Sinoquet, H. (2018). Assessment of radish (*Raphanus sativus* L.) response to fertilization based on plant growth analysis. *European Journal of Agronomy*, 92, 80-91.

Mikkelsen, R. L., Wilson, R. L., & Davis, K. (2016). *Soil test interpretations and fertilizer management for vegetables*. University of Idaho Extension.

Rajasekaran, R., Mekala, V., & Aranganathan, N. (2019). Postharvest management of vegetables: A review. *Journal of Scientific and Industrial Research*, 78(3), 143-151.

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Yang, C., Gao, J., & Li, Z. (2018). Sugar metabolism in harvested tomato fruit and the role of its enzymes in determining fruit quality. *Trends in Food Science & Technology*, 79, 155-162.

Zotarelli, L., Dukes, M. D., Morgan, K. T., Haman, D. Z., & Scholberg, J. M. (2018). Organic Amendments and Their Influence on Soil Properties: A Review. *Soil Science Society of America Journal*, 82(6), 1569-1581.

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Appendices

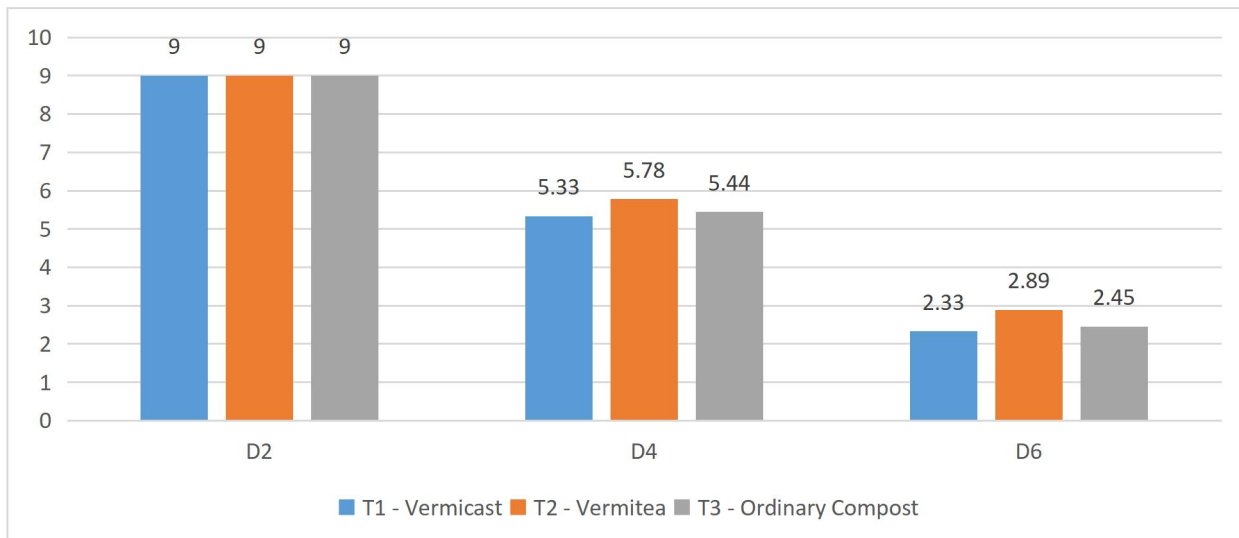


Figure 1. Visual Quality Rating (VQR) of kangkong applied with different soil ameliorant.

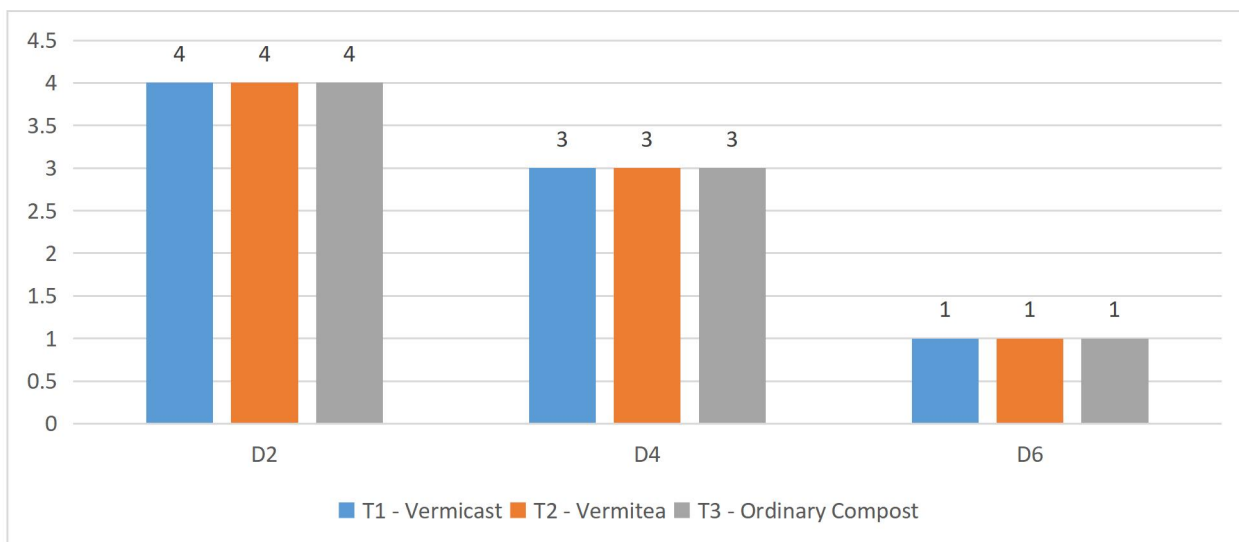


Figure 2. Shriveling rate of kangkong applied with different soil ameliorants.

TREATMENT	Initial pH ^{ns}	Final pH ^{ns}	Initial TSS (Brix) ^{ns}	Final TSS (Brix) ^{ns}
T1 - Vermicast	6.34	5.9	1.27	2.63
T2 - Vermitea	6.25	5.99	1.53	2.73
T3 - Ordinary Compost	6.3	5.88	1.9	2.9

Table 1. Initial pH, Final pH, Initial TSS and Final TSS of kangkong applied with different soil ameliorant.